Prelab 5

Sunday, March 22, 2026 7:28 PM

Name: Tridib Banik
St#: 400514461
MacID: banikt
Section: L03
Group: 7

Ans. to the Q. No. 1

The transfer function used in lab was:

$$G_c(s) = K_c(s+z) = K_c s + K_c z$$

Setting this equal to the first form:

$$K_D s + K_p = K_c s + K_c z$$

Matching coefficients:

coefficient of s: $K_c = K_D$ ---- (i)

constant terms: $K_c z = K_p$

$$\therefore z = \frac{K_p}{K_c}$$

$$\langle K_c = K_D \text{ from (i)} \rangle$$

$$\therefore z = \frac{K_p}{K_D}$$

∴ Solving for K_c and z in terms of K_D and K_p:

$$K_c = K_D$$

$$z = \frac{K_p}{K_D}$$

Ans. to the Q. No. 2(a)

The open-loop transfer function (controller × plant in cascade) is:

$$G_{openloop}(s) = K_c(s+z)\frac{0.419}{s^2}$$

This system has 2 open-loop poles at the origin.

Static error constant for position (position constant),

$$K_p = \lim_{s\to 0} G_{openloop}(s) = \frac{0.419K_c z}{0} = \infty$$

$$\therefore K_p = \infty$$

Ans. to the Q. No. 2(b)

Static error constant for velocity (velocity constant),

$$K_v = \lim_{s\to 0} s \cdot G_{openloop}(s) = \lim_{s\to 0} \frac{0.419K_c s(s+z)}{s^2}$$

$$= \lim_{s\to 0}\left(\frac{0.419K_c s^2}{s} + \frac{0.419K_c z}{s^2}\right)$$

$$= \lim_{s\to 0} 0.419K_c s + \lim_{s\to 0}\frac{0.419K_c z}{s^2}$$

$$= 0 + \infty$$

$$= \infty$$

$$\therefore K_v = \infty$$

Ans. to the Q. No. 2(c)

Static error constant for acceleration (acceleration constant),

$$K_a = \lim_{s\to 0} s^2 \cdot G_{openloop}(s) = \lim_{s\to 0} s^2 \cdot \frac{0.419K_c(s+z)}{s^2}$$

$$= \lim_{s\to 0} 0.419K_c(s+z)$$

$$= 0.419K_c z$$

$$\therefore K_a = 0.419K_c z$$

Ans. to the Q. No. 2(d)

Steady state error for a step input $u(t)$,

$$e_{ss} = \frac{1}{1+K_p} = \frac{1}{1+\infty}$$

$$\therefore e_{ss} = 0$$

Ans. to the Q. No. 2(e)

Steady state error for a ramp input $tu(t)$,

$$e_{ss} = \frac{1}{K_v} = \frac{1}{\infty}$$

$$\therefore e_{ss} = 0$$

Ans. to the Q. No. 2(f)

Steady state error for a parabolic input $t^2u(t)$,

$$e_{ss} = \frac{1}{K_a} = \frac{1}{0.419K_c z}$$

Substituting $K_c = K_D$ and $z = \frac{K_p}{K_D}$ from Q1:

$$e_{ss} = \frac{1}{0.419(K_D)\left(\frac{K_p}{K_D}\right)}$$

$$= \frac{1}{0.419K_p}$$

Ans. to the Q. No. 3(a)

The PID controller is:

$$G_c(s) = K_D s + K_p + \frac{K_I}{s}$$

$$= \frac{K_D s^2 + K_p s + K_I}{s}$$

The open-loop transfer function (cascade with plant) is:

$$G_{openloop}(s) = \frac{K_D s^2 + K_p s + K_I}{s} \cdot \frac{0.419}{s^2}$$

$$= \frac{0.419(K_D s^2 + K_p s + K_I)}{s^3}$$

The system has 3 poles at the origin.

Static error constant for position (position constant),

$$K_p = \lim_{s\to 0} G_{openloop}(s) = \frac{0.419(K_D(0)^2 + K_p(0) + K_I)}{0^3}$$

$$= \infty$$

$$\therefore K_p = \infty$$

Ans. to the Q. No. 3(b)

Static error constant for velocity (velocity constant),

$$K_v = \lim_{s\to 0} s \cdot G_{openloop}(s) = \lim_{s\to 0} s \cdot \frac{0.419(K_D s^2 + K_p s + K_I)}{s^3}$$

$$= \lim_{s\to 0} \frac{0.419(K_D s^2 + K_p s + K_I)}{s^2}$$

$$= \frac{0.419(K_D(0)^2 + K_p(0) + K_I)}{0^2}$$

$$= \infty$$

$$\therefore K_v = \infty$$

Ans. to the Q. No. 3(c)

Static error constant for acceleration (acceleration constant),

$$K_a = \lim_{s\to 0} s^2 \cdot G_{openloop}(s) = \lim_{s\to 0} s^2 \cdot \frac{0.419(K_D s^2 + K_p s + K_I)}{s^3}$$

$$= \lim_{s\to 0} \frac{0.419(K_D s^2 + K_p s + K_I)}{s}$$

$$= \frac{0.419(K_D(0)^2 + K_p(0) + K_I)}{0}$$

$$= \infty$$

$$\therefore K_a = \infty$$

Ans. to the Q. No. 3(d)

Steady state error for a step input $u(t)$,

$$e_{ss} = \frac{1}{1+K_p} = \frac{1}{1+\infty}$$

$$\therefore e_{ss} = 0$$

Ans. to the Q. No. 3(e)

Steady state error for a ramp input $tu(t)$,

$$e_{ss} = \frac{1}{K_v} = \frac{1}{\infty} = 0$$

$$\therefore e_{ss} = 0$$

Ans. to the Q. No. 3(f)

Steady state error for a parabolic input $t^2u(t)$,

$$e_{ss} = \frac{1}{K_a} = \frac{1}{\infty} = 0$$

$$\therefore e_{ss} = 0$$

Ans. to the Q. No. 4(a)

Since we ignore input R(s), $R(s) = 0$

$$E(s) = R(s) - C(s)$$

$$\langle R(s) = 0 \rangle$$

$$= 0 - C(s)$$

$$\therefore E(s) = -C(s)$$

From the block diagram,

$$\big(E(s)G_1(s) + D(s)\big)G_2(s) = C(s)$$

$$\langle E(s) = -C(s) \rangle$$

$$\big(-C(s)G_1(s) + D(s)\big)G_2(s) = C(s)$$

$$-C(s)G_1(s)G_2(s) + D(s)G_2(s) = C(s)$$

$$D(s)G_2(s) = C(s) + C(s)G_1(s)G_2(s)$$

$$D(s)G_2(s) = C(s)\big(1 + G_1(s)G_2(s)\big)$$

$$\frac{D(s)G_2(s)}{1 + G_1(s)G_2(s)} = C(s)$$

$$\therefore C(s) = \frac{G_2(s)}{1 + G_1(s)G_2(s)} D(s)$$

$$\langle E(s) = -C(s) \quad \therefore C(s) = -E(s) \rangle$$

$$E(s) = \frac{-G_2(s)}{1 + G_1(s)G_2(s)} D(s)$$

$$\therefore \frac{E(s)}{D(s)} = \frac{-G_2(s)}{1 + G_1(s)G_2(s)}$$

(Ans)

Ans. to the Q. No. 4(b)

$$G_1(s) = K_c(s+z)$$

$$G_2(s) = \frac{0.419}{s^2}$$

$$\therefore G_1(s)G_2(s) = \frac{0.419K_c(s+z)}{s^2}$$

Using superposition: Error due to R(s) alone (set $D(s) = 0$):

$$R(s) - C(s) = E(s) \Rightarrow R(s) = C(s) + E(s) \Rightarrow R(s) = E(s)G_1(s)G_2(s) + E(s)$$

$$\Rightarrow R(s) = E(s)\big(G_1(s)G_2(s) + 1\big)$$

$$\therefore \frac{R(s)}{E(s)} = \frac{G_1(s)G_2(s) + 1}{1}$$

$$\text{So, } \frac{E(s)}{R(s)} = \frac{1}{1 + G_1(s)G_2(s)}$$

$$\frac{E(s)}{R(s)} = \frac{1}{1 + G_1G_2} = \frac{1}{1 + \frac{0.419K_c(s+z)}{s^2}}$$

$$= \frac{1}{\frac{s^2 + 0.419K_c(s+z)}{s^2}}$$

$$= \frac{s^2}{s^2 + 0.419K_c(s+z)}$$

$$e_{ss}^R = \lim_{s\to 0} s \cdot \frac{s^2}{s^2 + 0.419K_c(s+z)} \cdot \frac{A}{s}$$

$$= \lim_{s\to 0} \frac{As^2}{s^2 + 0.419K_c(s+z)}$$

$$= \frac{A(0)^2}{0^2 + 0.419K_c(0+z)}$$

$$= 0$$

Error due to D(s) alone (set $R = 0$):

$$C(s) = \big(E(s)G_1(s) + D(s)\big)G_2(s)$$

$$-E(s) = E(s)G_1(s)G_2(s) + D(s)G_2(s)$$

$$-E(s)G_1(s)G_2(s) - E(s) = D(s)G_2(s)$$

$$-E(s)\big(G_1(s)G_2(s) + 1\big) = D(s)G_2(s)$$

$$\frac{E(s)}{D(s)} = \frac{-G_2(s)}{G_1(s)G_2(s) + 1}$$

$$= \frac{-\frac{0.419}{s^2}}{1 + K_c(s+z)\frac{0.419}{s^2}}$$

$$= \frac{-\frac{0.419}{s^2}}{\frac{s^2 + K_c(s+z)0.419}{s^2}}$$

$$= \frac{-0.419}{s^2 + K_c(s+z)0.419}$$

$$e_{ss}^D = \lim_{s\to 0} s \cdot \frac{-0.419}{s^2 + K_c(s+z)0.419} \cdot \frac{B}{s}$$

$$= \lim_{s\to 0} \frac{-0.419B}{s^2 + K_c(s+z)0.419}$$

$$= \frac{-0.419B}{0 + K_c(0+z)0.419}$$

$$= \frac{-0.419B}{K_c z(0.419)}$$

$$= \frac{-B}{K_c z}$$

∴ Total steady-state, $e_{ss} = 0 + \frac{-B}{K_c z} = \frac{-B}{K_c z}$

∴ the PD controller can't reject the step disturbance; a non-zero steady-state error remains due to D(s).

Ans. to the Q. No. 4(c)

$$G_1(s) = K_D s + K_p + \frac{K_I}{s}$$

$$= \frac{K_D s^2 + K_p s + K_I}{s}$$

$$G_2(s) = \frac{0.419}{s^2}$$

$$G_1G_2 = \frac{0.419(K_D s^2 + K_p s + K_I)}{s^3}$$

Error due to R(s) alone ($D(s) = 0$)

$$E(s) = R(s) - C(s)$$

$$E(s) = R(s) - \big(E(s)G_1(s)G_2(s)\big)$$

$$E(s) + E(s)G_1(s)G_2(s) = R(s)$$

$$E(s)\big(1 + G_1(s)G_2(s)\big) = R(s)$$

$$E(s) = \frac{R(s)}{1 + G_1(s)G_2(s)}$$

$$\frac{E(s)}{R(s)} = \frac{1}{1 + G_1(s)G_2(s)}$$

$$= \frac{1}{1 + \frac{0.419(K_D s^2 + K_p s + K_I)}{s^3}}$$

$$= \frac{1}{\frac{s^3 + 0.419(K_D s^2 + K_p s + K_I)}{s^3}}$$

$$= \frac{s^3}{s^3 + 0.419(K_D s^2 + K_p s + K_I)}$$

$$e_{ss}^R = \lim_{s\to 0} s \cdot \frac{s^3}{s^3 + 0.419(K_D s^2 + K_p s + K_I)} \cdot \frac{A}{s}$$

$$= \frac{A(0)^3}{(0)^3 + 0.419(K_D(0)^2 + K_p(0) + K_I)}$$

$$= 0$$

E(s) due to D(s) alone ($R(s) = 0$):

$$R(s) - C(s) = E(s)$$

$$0 - C(s) = E(s)$$

$$-\Big(\big(E(s)G_1(s) + D(s)\big)G_2(s)\Big) = E(s)$$

$$-E(s)G_1(s)G_2(s) - D(s)G_2(s) = E(s)$$

$$D(s)G_2(s) = E(s) + E(s)G_1(s)G_2(s)$$

$$D(s)G_2(s) = E(s)\big(1 + G_1(s)G_2(s)\big)$$

$$\frac{E(s)}{D(s)} = \frac{G_2(s)}{1 + G_1(s)G_2(s)}$$

$$\frac{E(s)}{D(s)} = \frac{\frac{0.419}{s^2}}{1 + \frac{0.419(K_D s^2 + K_p s + K_I)}{s^3}}$$

$$= \frac{\frac{0.419}{s^2}}{\frac{s^3 + 0.419(K_D s^2 + K_p s + K_I)}{s^3}}$$

$$= \frac{0.419}{s^2} \cdot \frac{s^3}{s^3 + 0.419(K_D s^2 + K_p s + K_I)}$$

$$= \frac{0.419s}{s^3 + 0.419(K_D s^2 + K_p s + K_I)}$$

$$e_{ss}^D = \lim_{s\to 0} s \cdot \frac{0.419s}{s^3 + 0.419(K_D s^2 + K_p s + K_I)} \cdot \frac{B}{s}$$

$$= \frac{0.419(0)}{(0)^3 + 0.419(K_D(0)^2 + K_p(0) + K_I)}$$

$$= 0$$

∴ Total steady-state error: $e_{ss} = 0$

∴ The integrator in the PID controller completely rejects the step disturbance.